Chapter 7

Chemical Equilibrium

Learning Objectives

- Explain chemical equilibrium
- Write an equilibrium equation and the expression for its equilibrium constant K_c (or K_p)
- Calculate K_c (or K_p) or equilibrium concentration for a given reaction

- Define Le Chatelier's principle
- Apply Le Chatelier's principle to predict the position of equilibrium for a reversible reaction with changes in concentration, pressure, volume or temperature

- Calculate the pH of acid, base and buffer solution

Part 1

Equilibrium and Equilibrium Constant

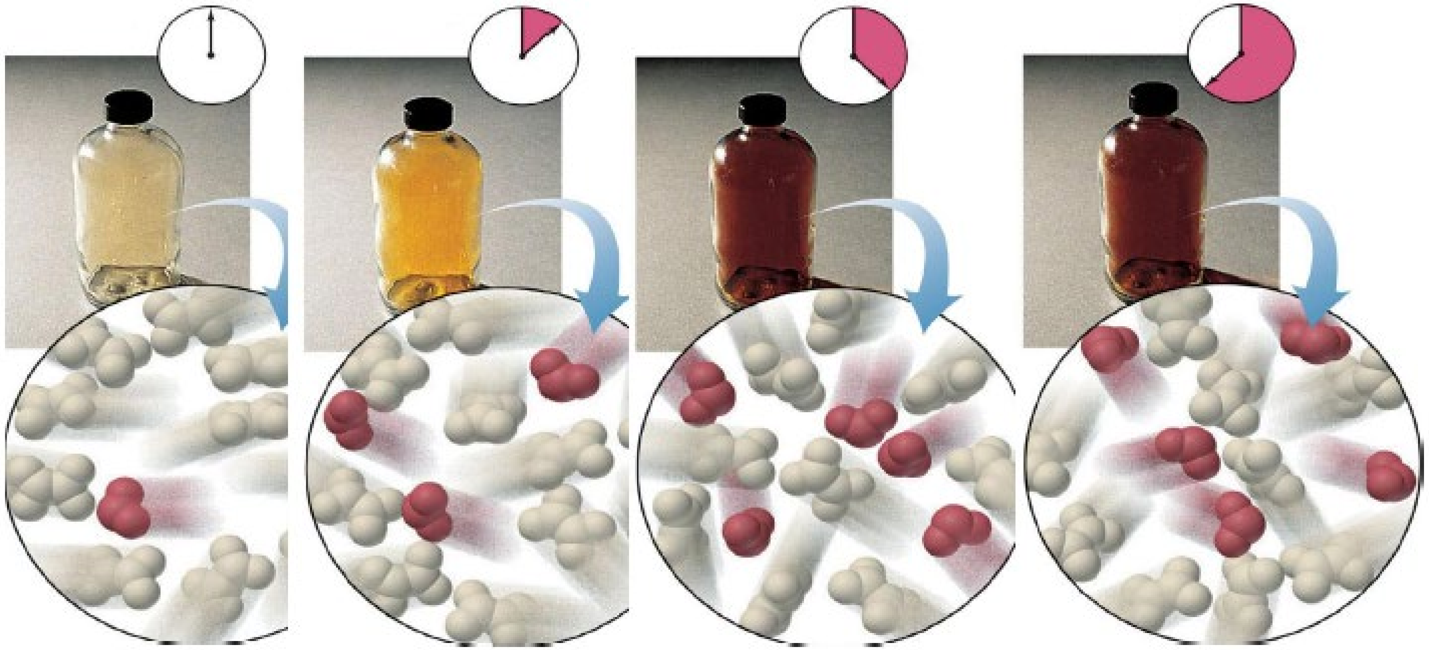
Kinetics vs Equilibrium

- Kinetics ("how fast")
  - is concerned with the *rate* of a reaction
  - the concentration of the product that appears (or of reactant that disappears) per unit time

- Equilibrium ("how complete")
  - is concerned with the *extent* (or yield) of a reaction
  - the concentrations of reactant and product present after infinite time or once no further changes occur

Example: Dissociation of N_2O_4 to NO_2

N_2O_4 (g, colourless) $\rightleftharpoons$ $2NO_2$ (g, brown)



At equilibrium: $\text{rate}_{\text{forward}} = \text{rate}_{\text{reverse}}$

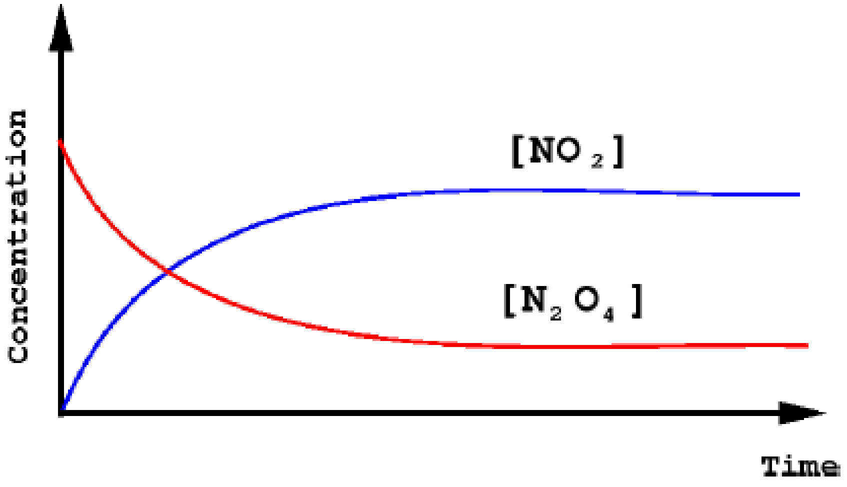
1 mole of A is converted to 2 moles of B per second
2 moles of B is converted to 1 mole of A per second

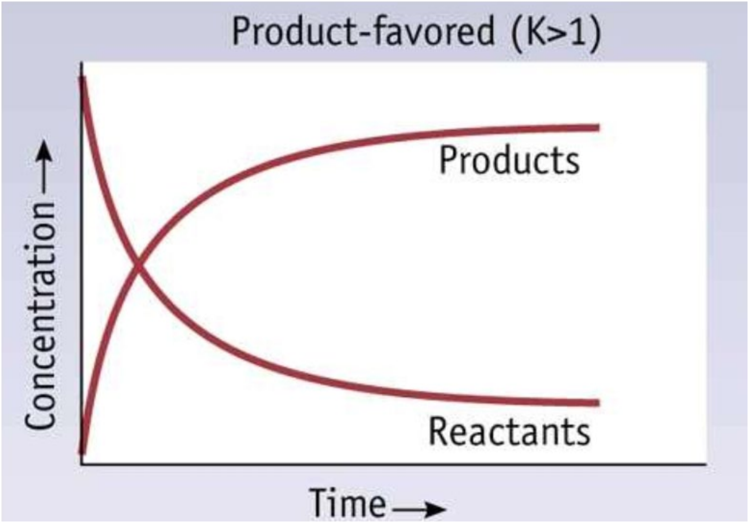
Equilibrium Expression for $N_2O_4 \rightleftharpoons NO_2$

$$N_2O_4(g) \rightleftharpoons 2NO_2(g)$$

$$k_{forward}[N_2O_4] = k_{reverse}[NO_2]^2$$

$$\frac{k_{forward}}{k_{reverse}} = \frac{[NO_2]^2_{eq}}{[N_2O_4]_{eq}} = K$$





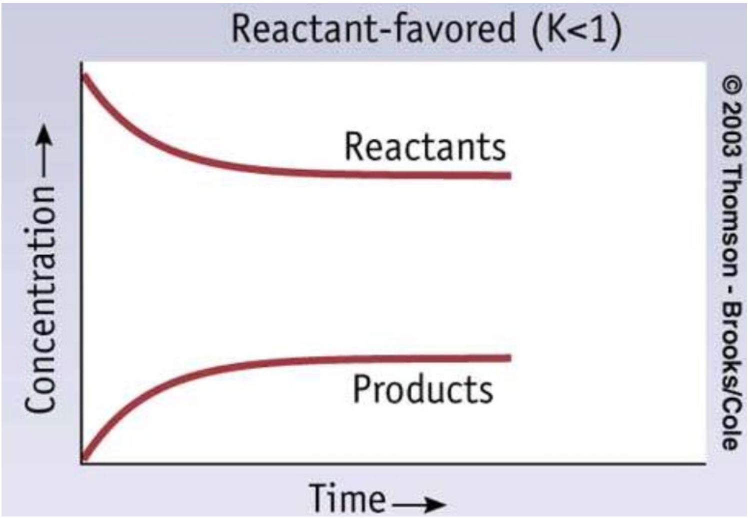
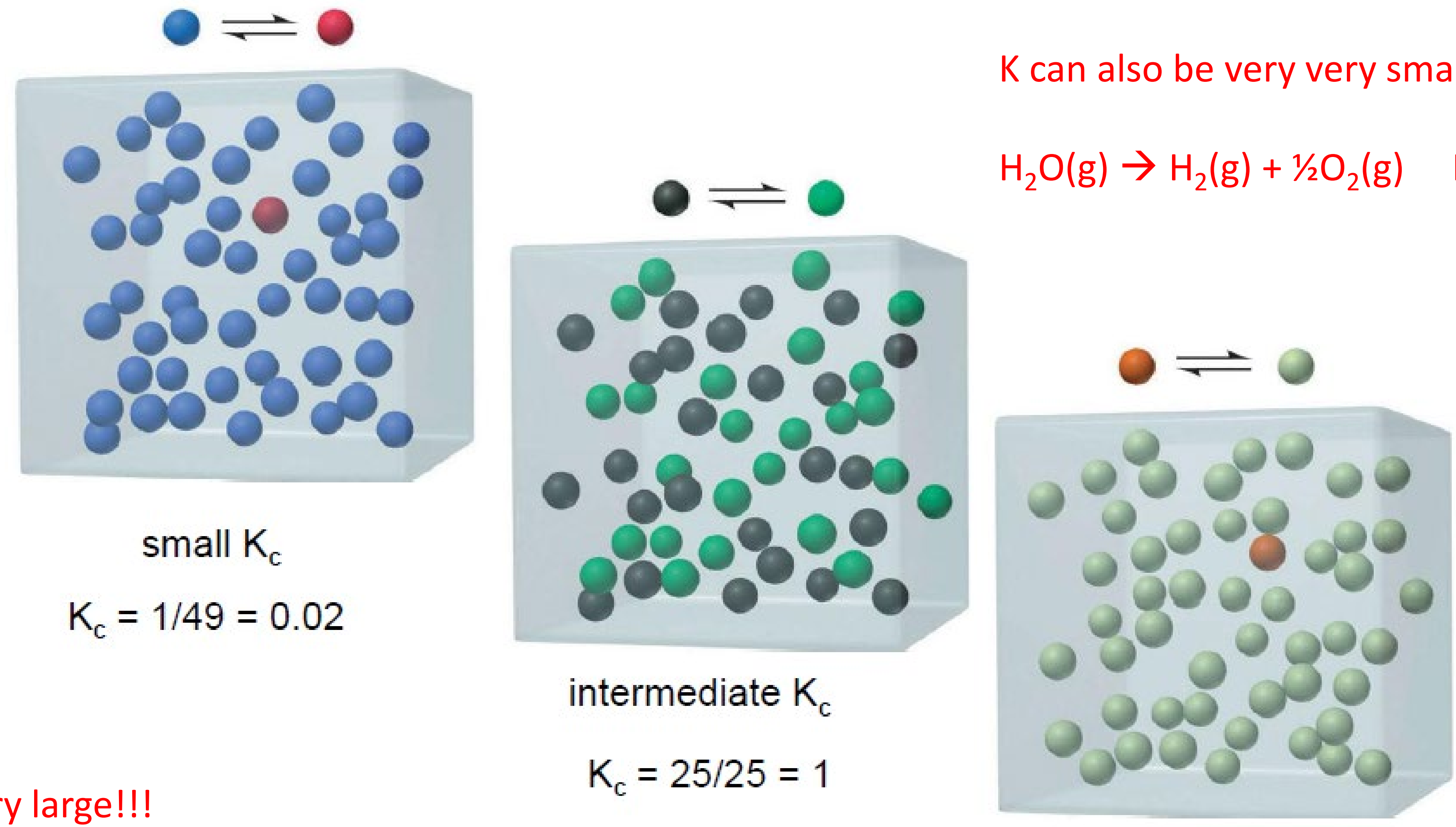


Figure 17.2 The **range** of equilibrium constants

K can also be very very small!!!

$H_2O(g) \rightarrow H_2(g) + ½O_2(g)$ $K(298K) = 8.6*10^{-41}$



small K_c

$K_c = 1/49 = 0.02$

intermediate K_c

$K_c = 25/25 = 1$

large K_c

$K_c = 49/1 = 49$

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K can be very very large!!!

$H_2(g) + ½O_2(g) \rightarrow H_2O(g)$ $K(298K) = 1.2*10^{40}$

Equilibrium constant K_c , K_p *and K*

- Independent of concentration and pressure
- Dependent of temperature → for each different T, there is one unique K → K = K(T)
- It tells us the extent (yield) of a reaction and the final composition of the reaction mixture, after infinite time (in practice, sufficiently long until no further change occurs).
- A system at equilibrium is dynamic on the molecular level: no further net change is observed because changes in one direction are balanced by changes in the other.
- Concentration of reactants and products becomes constant (equilibrium concentration), but not necessarily equal.
- In general, products are favored if $K > 1$ or $K >> 1$; reactants are favored if $K < 1$ or $K << 1$.

Expression of Equilibrium Constant

- For a generic reaction at equilibrium

$$aA + bB \rightleftharpoons cC + dD$$

$$K_C = \frac{[C]^c[D]^d}{[A]^a[B]^b}$$

- For the reverse reaction:

$$cC + dD \rightleftharpoons aA + bB$$

$$K'_C = \frac{[A]^a[B]^b}{[C]^c[D]^d}$$

$$= \frac{1}{K_C}$$

Exercise: write the K_c, K_p, K expression of:

$$4NH_3(g) + 3O_2(g) \rightleftharpoons 2N_2(g) + 6H_2O(g)$$

- Note: the products always appear in the numerator and the reactants in the denominator.

Equilibrium Constants of "combined reaction"

Multiple equilibria

$$N_2(g) + O_2(g) \leftrightharpoons \cancel{2NO(g)} \qquad K_c(1) = \frac{[NO]^2}{[N_2][O_2]} = 2.3 \times 10^{-19}$$

$$\cancel{2NO(g)} + O_2(g) \leftrightharpoons 2NO_2(g) \qquad K_c(2) = \frac{[NO_2]^2}{[NO]^2[O_2]} = 3 \times 10^{6}$$

$$N_2(g) + 2O_2(g) \leftrightharpoons 2NO_2(g) \qquad K_c(\text{overall}) = \frac{[NO_2]^2}{[N_2][O_2]^2} = K_c(1) \times K_c(2)$$

$$= 6.9 \times 10^{-13}$$

Reactions involving Solids and Liquids

- Solids and liquids are not included in *K*
  - this is because their concentration is essentially unchanged by the reaction
  - ie. concentrations of any solid reactants and products, the molar concentration of water (or of any liquid reactant or product) are omitted from the equilibrium constant expression
- For example:

$$\frac{1}{8}S_8(s) + O_2(g) \rightleftharpoons SO_2(g)$$

$$\text{Initially, we write :} \qquad K'_C = \frac{[SO_2]}{\cancel{[S_8]^{\frac{1}{8}}}[O_2]}$$

$$\text{But, in fact :} \qquad K_C = \frac{[SO_2]}{[O_2]}$$

Exercise: write K for each reaction:

1. $CH_3COOH(aq) + H_2O(l) \rightleftharpoons CH_3COO^-(aq) + H_3O^+(aq)$
2. $Hg(l) + Cl_2(g) \rightleftharpoons HgCl_2(s)$
3. $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

Unit of Equilibrium Constant

- The true IUPAC definition of K is:

$$K_C = \frac{([C]/[C]^\circ)^c([D]/[D]^\circ)^d}{([A]/[A]^\circ)^a([B]/[B]^\circ)^b} \qquad K_P = \frac{(P_C/P_C^\circ)^c(P_D/P_D^\circ)^d}{(P_A/P_A^\circ)^a(P_B/P_B^\circ)^b}$$

- $[A]^o$, $[B]^o$, … are reference concentration of standard state, i.e. 1 mol dm^{-3} → unit of [A] and $[A]^o$ cancel out → K_c is dimensionless
- Likewise, standard state p^o = 1 bar (NOT 1 atm or 1 Pa) → when unit of p_A, p_B, … is bar, the unit cancel out → K_p is also dimensionless

Reaction Quotient (Q)

- Reaction may not always be at equilibrium. In fact, most of the time, the reaction is NOT at equilibrium.
- At any time (equilibrium or non-equilibrium), reaction quotient can be calculated:

$NH_4^+ + CH_3COO^- \leftrightharpoons NH_3 + CH_3COOH$ (in water)

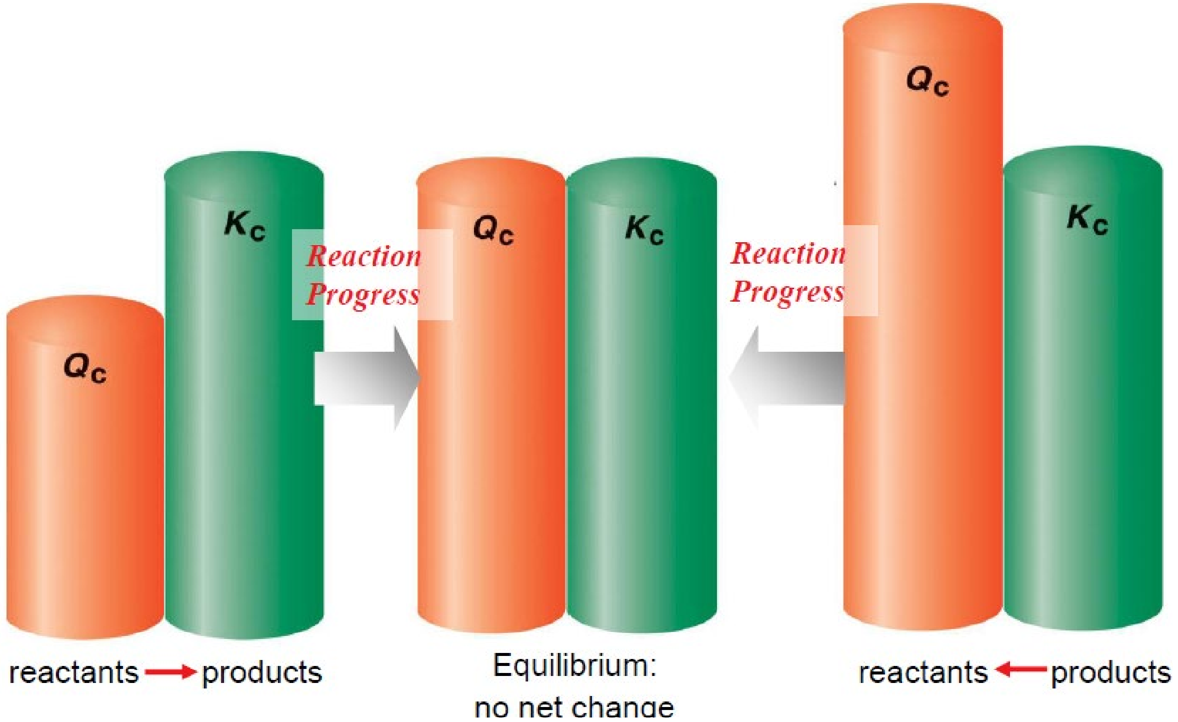
$$Q_c = \frac{[NH_3][CH_3COOH]}{[NH_4^+][CH_3COO^-]}$$ (non equilibrium system)

$$K_c = \frac{[NH_3][CH_3COOH]}{[NH_4^+][CH_3COO^-]}$$ (equilibrium system)

The two expressions look identical → the only difference is whether the concentration is equilibrium concentration or not

Q and direction of reaction

- Q > K: [more products & less reactants] reverse reaction predominant
- Q = K: reaction is at equilibrium
- Q < K: [less product & more reactants] forward reaction predominant



Equilibrium calculation: ICE table (1)

H_2 (g) + I_2 (g) ⇌ 2HI (g)

Begin with 1 mol L^{-1} of HI, and let reaction reach equilibrium with H_2 and I_2 at 25°C (**K_c = 25**). What will be the final concentrations?

	H_2 (g)	+	I_2 (g)	⇌	2HI (g)
Initial (M)	0		0		1
Change (M)	$+x$		$+x$		$-2x$
Equilibrium (M)	x		x		$1-2x$

How about starting from 0.5 mol L^{-1} of H_2 and 0.5 mol L^{-1} of I_2?

It is a good habit to **confirm** your final answer is indeed the "equilibrium concentration"

Equilibrium calculation: ICE table (2)

$PCl_5\ (g) \rightleftharpoons PCl_3\ (g) + Cl_2\ (g)$

When a flask that initially contained only pure PCl_5 was heated to 250 ^{o}C, 68.6% of the PCl_5 was dissociated at equilibrium. The total pressure of all gases at equilibrium was 2.00 bar. Find *K*.

	$PCl_5\ (g)$	$\rightleftharpoons$	$PCl_3\ (g)$	+	$Cl_2\ (g)$
Initial (mol)					
Change (mol)					
Eqm (mol)					

Part 2

Le Chatelier's Principle

Le Chatelier's Principle

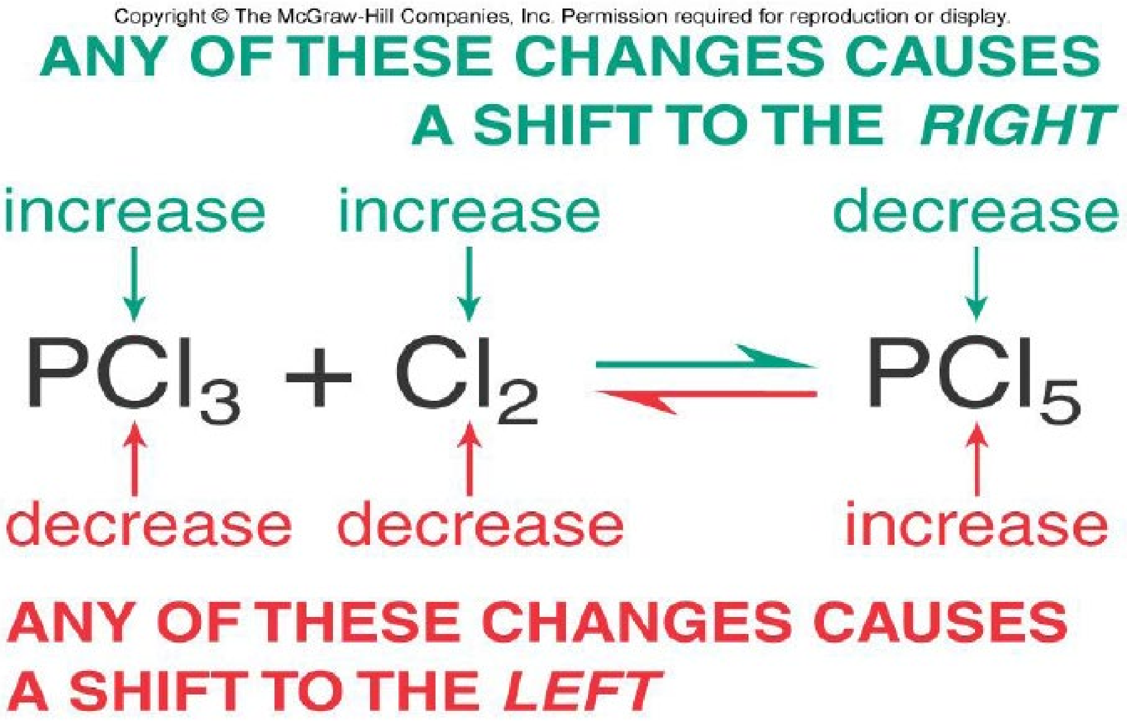
"If the equilibrium of a system is disturbed by a change, the system tends to adjust itself to a new equilibrium by counteracting of the change."

Simple version: equilibrium will shift to counteract the disturbance

- There are 4 types of disturbance:
  - amount of reactants and/or products
  - pressure
  - temperature
  - catalyst

Counteract

Le Chatelier's Principle – concentration

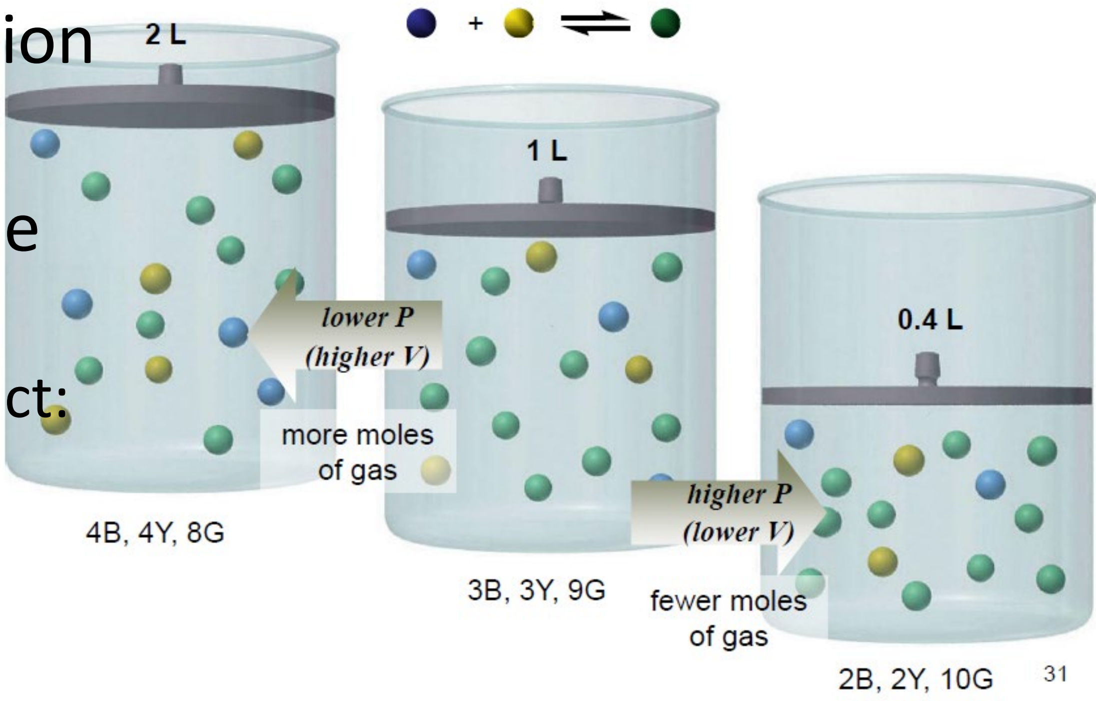


- Adding product makes Q > K $\longleftarrow$
- Removing reactant makes Q > K $\longleftarrow$
- Adding reactant makes Q < K $\longrightarrow$
- Removing product makes Q < K $\longrightarrow$

- Pressure of individual component has the same effect as concentration: $pV = nRT \rightarrow p/RT = n/V = c \rightarrow$ c is directly proportional to p (at constant T)

Le Chatelier's Principle – (total) pressure

- Pressure only affect gas, but not solution (or liquid/solid)
- Change partial pressure of individual component has the same effect as individual concentration
- Change the total pressure of the system (typically by compression or expansion of the total volume)
  - V(total) ↓ → p(total) ↑ → in order to counteract: shift to decrease p → shift to the side with less gaseous molecules



Le Chatelier's Principle – temperature

- When c / p / V changes, Q will change, but K remains constant
- Only when T changes, K will change [K = K(T)]
- If forward reaction is exothermic (ΔH = –ve) then reverse reaction is endothermic (ΔH = +ve).
- For exothermic reaction: treat heat as a "product" $aA + bB \rightleftharpoons cC + dD + \text{heat}$
  - T $\uparrow$ $\rightarrow$ "product" (heat) $\uparrow$ $\rightarrow$ to counteract, equilibrium shift towards left
  - T $\downarrow$ $\rightarrow$ "product" (heat) $\downarrow$ $\rightarrow$ to counteract, equilibrium shift towards right
- For endothermic reaction: treat heat as a "reactant" $aA + bB + \text{heat} \rightleftharpoons cC + dD$
  - T $\uparrow$ $\rightarrow$ "reactant" (heat) $\uparrow$ $\rightarrow$ to counteract, equilibrium shift towards right
  - T $\downarrow$ $\rightarrow$ "reactant" (heat) $\downarrow$ $\rightarrow$ to counteract, equilibrium shift towards left

Table 17.4 Effect of Various Disturbances on an Equilibrium System

Disturbance	Net Direction of Reaction	Effect on Value of *K*
Concentration		
Increase [reactant]	Toward formation of product	None
Decrease [reactant]	Toward formation of reactant	None
Increase [product]	Toward formation of reactant	None
Decrease [product]	Toward formation of product	None
Pressure		
Increase *P* (decrease *V*)	Toward formation of fewer moles of gas	None
Decrease *P* (increase *V*)	Toward formation of more moles of gas	None
Increase *P* (add inert gas, no change in *V*)	None; concentrations unchanged	None
Temperature		
Increase *T*	Toward absorption of heat	Increases if $\Delta H^0_{rxn} > 0$ Decreases if $\Delta H^0_{rxn} < 0$
Decrease *T*	Toward release of heat	Increases if $\Delta H^0_{rxn} < 0$ Decreases if $\Delta H^0_{rxn} > 0$
Catalyst added	None; forward and reverse equilibrium attained sooner; rates increase equally	None

The Industrial Production of Ammonia

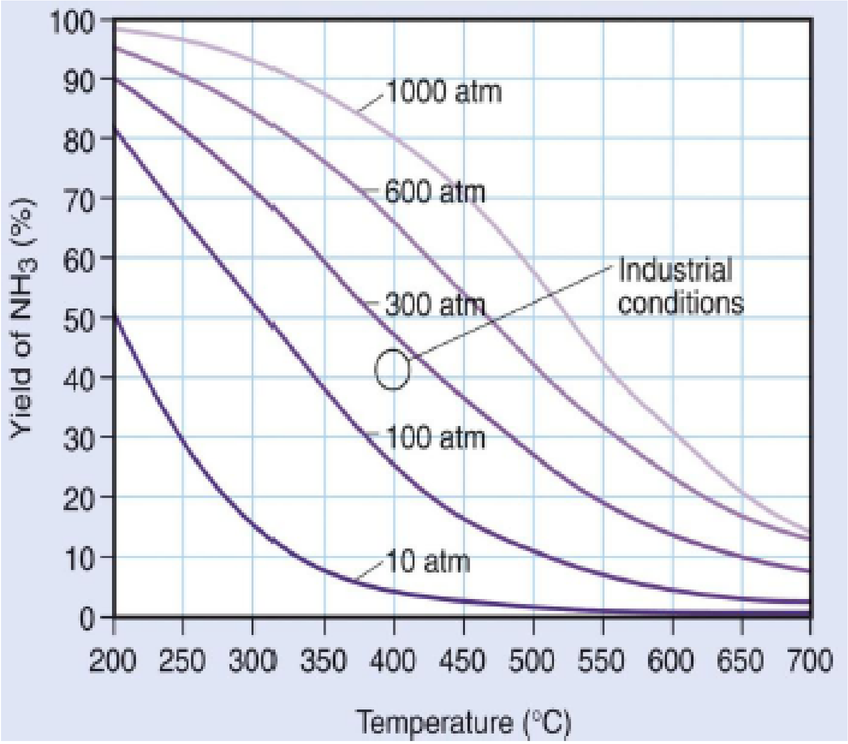
Ammonia is prepared industrially through the Haber process. There are three ways to maximize the yield:

1. Decrease concentration of ammonia:
   Ammonia is a product, so removing it will shift the equilibrium to the right.

2. Increase pressure:
   Since 4 moles of gas react to form 2 moles of gas, decreasing the volume will shift the equilibrium to right.

3. Decrease in temperature:
   Formation of ammonia is exothermic, so decreasing the temperature will shift the equilibrium to right.

$$N_2\,(g) + 3H_2\,(g) \rightleftharpoons 2NH_3\,(g)$$



Percent yield of ammonia vs. temperature (°C) at five different operating pressures.

Equilibrium vs Kinetics

- Although yield is favored at this low temperature, the rate of formation is too slow to be economical.
- In practice, a compromise optimizes both yield and rate.
- High pressure and continuous removal of product increase yield.
- The slightly higher temperature and a catalyst are used to increase rate.
- Modern ammonia plants operate at 200 – 300 atm and around 673K
- And the catalyst consist of small iron crystal fused into a mixture of MgO, Al_2O_3 and SiO_2.

Part 3

Acid-Base Equilibrium

Acid and Base

	Acid	Base	Property
Arrhenius definition	In aqueous solution, dissociate to form H^+ HCl, H_2SO_4, CH_3COOH	In aqueous solution, dissociate to form OH^- $NaOH$, KOH, $Ca(OH)_2$	Limited to aqueous solution
Bronsted-Lowry definition	Proton (H^+) donor H_4SiO_4, NH_4Cl	Proton (H^+) acceptor MgO, KH, $NaOCH_3$	No requirement of solvent. Definition works for other solvents, or even without solvent
Lewis definition	Electron pair acceptor BF_3, $FeCl_3$, $AlCl_3$	Electron pair donor $(CH_3CH_2)_2O$, CO	Always a pair of electrons, not a single electron

Conjugate Acid-Base pair

Conjugate Pair

Acid + Base $\rightleftharpoons$ Base + Acid

Conjugate Pair

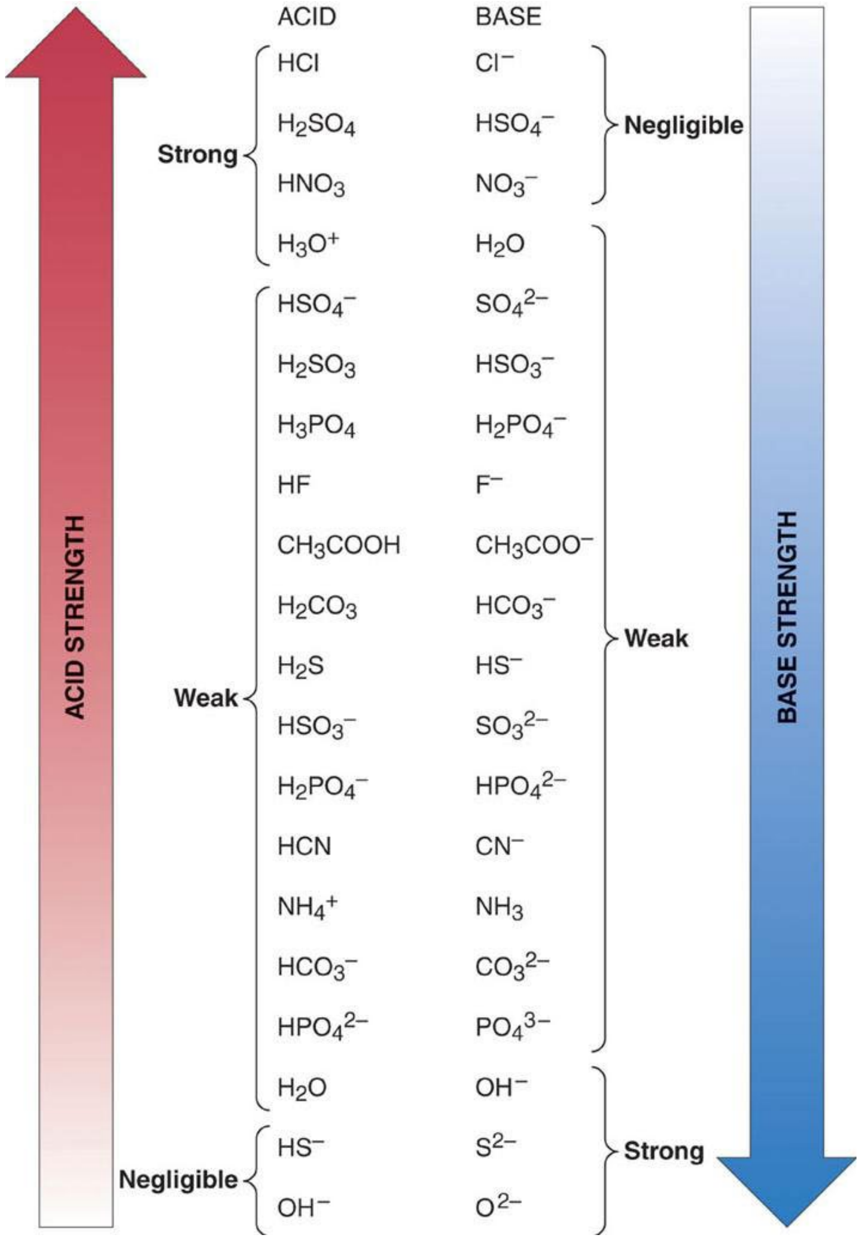
Reaction 1	HF	+	H_2O	$\rightleftharpoons$	F^-	+	H_3O^+
Reaction 2	$HCOOH$	+	CN^-	$\rightleftharpoons$	$HCOO^-$	+	HCN
Reaction 3	NH_4^+	+	CO_3^{2-}	$\rightleftharpoons$	NH_3	+	HCO_3^-
Reaction 4	$H_2PO_4^-$	+	OH^-	$\rightleftharpoons$	HPO_4^{2-}	+	H_2O
Reaction 5	H_2SO_4	+	$N_2H_5^+$	$\rightleftharpoons$	HSO_4^-	+	$N_2H_6^{2+}$
Reaction 6	HPO_4^{2-}	+	SO_3^{2-}	$\rightleftharpoons$	PO_4^{3-}	+	HSO_3^-

Strengths of conjugate acid-base pairs

- The stronger the acid, the weaker its conjugate base

- Strong acid vs weak acid

- Stronger acid vs weaker acid



The pH Scale

1. Provides a convenient way of expressing the hydrogen ion (H^+) concentration of a solution and describes acidity or basicity in quantitative terms.

2. $pH = -\log [H^+]$

3. Hydrogen ion concentration in pure water is
   1×10^{-7} *M at 25 °C [why?];*
   the pH of pure water is
   $-\log [1.0 \times 10^{-7}] = 7.00$

4. pH scale runs from pH = 0 to 14

5. Relationships between acidity, basicity, and pH:

   If pH = 7.0, the solution is neutral.

   If pH < 7.0, the solution is acidic.

   If pH > 7.0, the solution is basic.

Ionic product of water
$K_w = [H^+][OH^-] = 10^{-14}$ (25 °C)

Valid regardless whether the solution is acidic, basic or neutral.

Acid dissociation constant, K_a

- To measure and compare the relative strength of weak acids
- For monoprotic acid: HA $\leftarrow\rightarrow$ H^+ + A^-

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

- For diprotic acid:

$$H_2A \leftarrow\rightarrow H^+ + HA^- \quad K_{a,1} = \frac{[H^+][HA^-]}{[H_2A]}$$

$$HA^- \leftarrow\rightarrow H^+ + A^{2-} \quad K_{a,2} = \frac{[H^+][A^{2-}]}{[HA^-]}$$

- K_a is sometimes quite small (same as $[H^+]$), so pK_a is often used

$$pK_a = -\log(K_a)$$

pK_a table

From “Physical Chemistry” by Atkins

Table 7.4 Acidity constants for aqueous solutions at 298 K. (a) In order of acid strength

Acid	HA	A^-	K_a	pK_a
Hydriodic	HI	I^-	10^{11}	−11
Hydrobromic	HBr	Br^-	10^9	−9
Hydrochloric	HCl	Cl^-	10^7	−7
Sulfuric	H_2SO_4	HSO_4^-	10^2	−2
Perchloric*	$HClO_4$	ClO_4^-	4.0×10^1	−1.6
Hydronium ion	H_3O^+	H_2O	1	0.0
Oxalic	$(COOH)_2$	$HOOCCO_2^-$	5.6×10^{-2}	1.25
Sulfurous	H_2SO_3	HSO_3^-	1.4×10^{-2}	1.85
Hydrogensulfate ion	HSO_4^-	SO_4^{2-}	1.0×10^{-2}	1.99
Phosphoric	H_3PO_4	$H_2PO_4^-$	6.9×10^{-3}	2.16
Glycinium ion	$^+NH_3CH_2COOH$	NH_2CH_2COOH	4.5×10^{-3}	2.35
Hydrofluoric	HF	F^-	6.3×10^{-4}	3.20
Formic	$HCOOH$	HCO_2^-	1.8×10^{-4}	3.75
Hydrogenoxalate ion	$HOOCCO_2^-$	$C_2O_4^{2-}$	1.5×10^{-5}	3.81
Lactic	$CH_3CH(OH)COOH$	$CH_3CH(OH)CO_2^-$	1.4×10^{-4}	3.86
Acetic (ethanoic)	CH_3COOH	$CH_3CO_2^-$	1.4×10^{-5}	4.76
Butanoic	$CH_3CH_2CH_2COOH$	$CH_3CH_2CH_2CO_2^-$	1.5×10^{-5}	4.83
Propanoic	CH_3CH_2COOH	$CH_3CH_2CO_2^-$	1.4×10^{-5}	4.87
Anilinium ion	$C_6H_5NH_3^+$	$C_6H_5NH_2$	1.3×10^{-5}	4.87
Pyridinium ion	$C_5H_5NH^+$	C_6H_5N	5.9×10^{-6}	5.23
Carbonic	H_2CO_3	HCO_3^-	4.5×10^{-7}	6.35
Hydrosulfuric	H_2S	HS^-	8.9×10^{-8}	7.05
Dihydrogenphosphate ion	$H_2PO_4^-$	HPO_4^{2-}	6.2×10^{-8}	7.21
Hypochlorous	$HClO$	ClO^-	4.0×10^{-8}	7.40

pH calculation (1)

- Calculate the pH of the following solution:
- 0.2 mol dm^{-3} HCl
- 0.2 mol dm^{-3} H_2SO_4
- 0.2 mol dm^{-3} KOH
- 0.2 mol dm^{-3} $Ca(OH)_2$

pH calculation (2)

- Calculate the pH of the following solution:
- 0.2 mol dm^{-3} CH_3COOH (pK_a = 4.76)

- 0.2 mol dm^{-3} H_3PO_4 (pK_a = 2.16, 7.20, 12.32)

pH calculation (3)

- Calculate the pH of the following solution:
- 0.2 mol dm^{-3} NH_4Cl (pK_a of NH_4^+ = 9.25)

- 0.2 mol dm^{-3} NH_3 (pK_a of NH_4^+ = 9.25)

pH calculation (4)

- Ethanoic acid (CH_3COOH, 0.2 mol dm^{-3}, 100 cm^3) and sodium ethanoate (CH_3COONa, 0.2 mol dm^{-3}, 100 cm^3) are mixed, giving solution A.
- Calculate the "analytical concentration" of CH_3COOH and CH_3COONa. For simplicity, you can present them as HOAc and NaOAc (AcO^-).

- Calculate the pH of the solution A, using pK_a = 4.76.

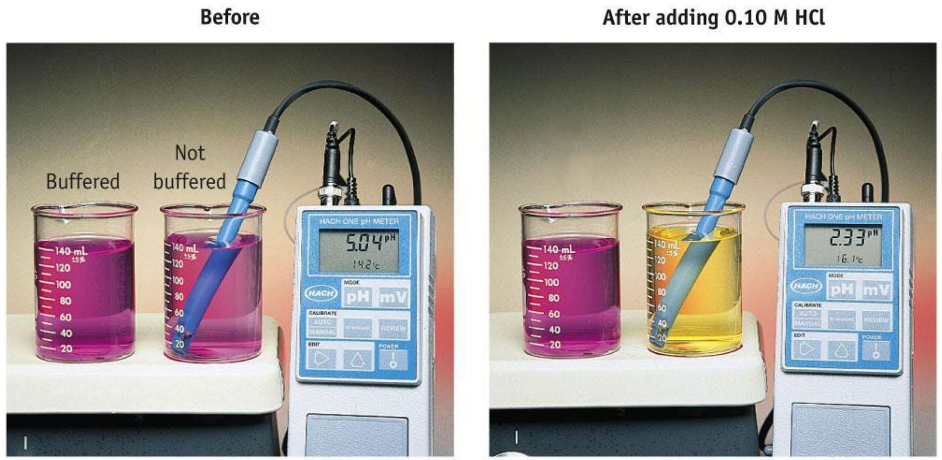
pH calculation (5)

- Calculate the pH of the obtained solutions:
- 5 cm^3 of 0.5 mol dm^{-3} NaOH is added to solution A

- 5 cm^3 of 0.5 mol dm^{-3} HCl is added to solution A

Buffer Solution

- A buffer causes solutions to be resistant to a change in pH when a strong acid or base is added.
- The components of a buffer are a weak acid, capable of reacting with added OH- ions and its conjugate base that can consume added H_3O^+ ions
- The acid and base must not react with each other.
- Eg: Acetic acid/ acetate ion buffer; NH_4Cl – NH_3 buffer; H2PO4- - HPO42- buffer
- Buffer range: pK_a–1 ~ pK_a+1; correct choice of buffer depends on pH requirement



(a) The pH electrode is indicating the pH of water that contains a trace of acid (and bromphenol blue acid–base indicator). The solution at the left is a buffer solution with a pH of about 7. (It also contains bromphenol blue dye.)

(b) When 5 mL of 0.10 M HCl is added to each solution, the pH of the water drops several units, whereas the pH of the buffer stays constant, as implied by the fact that the indicator color did not change.

Blood is a buffer that maintains constant pH with countless cellular acid base reactions